

IAL Physics Unit 2 (WPH12): Rapid Revision

Last-minute summary · Waves and Electricity · 1 h 30 min · 80 marks · 10 MCQ + structured + one 6-mark answer

EXAM RULES THAT COST MARKS

- “Show that”: 1 more s.f. than printed (4.43 eV, not 4.4 eV). If stuck, use the printed value next.
- Missing or wrong unit loses the final mark. Convert mm, nm, μs , minutes, kWh ($\times 3.6 \times 10^6$) and eV ($\times 1.60 \times 10^{-19}$) first.
- Radius, not diameter:** $A = \pi d^2/4$. The commonest lost mark in resistivity and drift velocity.
- “Is the student correct?”: your number, a comparison ($0.94 \text{ V} > 0.90 \text{ V}$), a conclusion. ecf applies.
- Refused words: “atoms” for electrons in energy levels; “bending” for refraction; “out of phase” for destructive (say **antiphase**); “size” for amplitude.
- 6-marker: 6 linked points for 6/6, joined with “so”, “because”, “therefore”.

CHARGE, CURRENT, P.D., POWER

$I = \Delta Q/\Delta t$	electrons = $Q/e = It/e$
$V = W/Q$	energy transferred per unit charge
$P = VI = I^2R = V^2/R$; $W = VIt$	same $V \rightarrow V^2/R$; same $I \rightarrow I^2R$
$R = V/I$	at a point on I-V curve; not the gradient

- e.m.f.:** energy supplied (work done) per unit charge by the source; = terminal p.d. when no current flows.

Ohmic resistor	straight line through origin
Filament lamp	V up $\rightarrow$ I up $\rightarrow$ T up $\rightarrow$ ions vibrate more $\rightarrow$ more collisions $\rightarrow$ R up
Thermistor (NTC)	T up $\rightarrow$ more conduction electrons (outweighs vibration) $\rightarrow$ R down
Diode / LED	no current in reverse or below threshold (0.7 V Si; $\sim 2 \text{ V LED}$)
LDR	more light $\rightarrow$ more photons $\rightarrow$ more conduction electrons $\rightarrow$ R down

CIRCUITS

- Series:** same I (conservation of charge); p.d.s add to e.m.f. (conservation of energy); $R_T = R_1 + R_2$.
- Parallel:** same p.d.; currents add at a junction; $1/R_T = 1/R_1 + 1/R_2$ (less than the smallest).
- Derivation:** $V = V_1 + V_2 \rightarrow IR_T = IR_1 + IR_2, \div I$. Parallel: $I = I_1 + I_2, \div V$. Name the conservation law.
- Solar cells: series $\rightarrow$ same I, larger V; parallel $\rightarrow$ same V, larger I, keeps working if one fails.

$V_{\text{out}} = V_{\text{in}} \times R_2/(R_1 + R_2)$	$V_1/V_2 = R_1/R_2$
Uniform wire	$V \propto \text{length: } V = V_{\text{total}} \times x/L$

“WHAT HAPPENS TO THE METER READINGS?”

- State the sensor’s R change (T or light up $\rightarrow$ R down).
- Total R changes, so I changes the opposite way.
- Sensor takes a smaller (larger) share of p.d.; fixed resistor the other way.
- Conclude for each meter. Never contradict yourself.

E.M.F. AND INTERNAL RESISTANCE

$\varepsilon = I(R + r) = V + Ir$	Ir = lost volts
$V = -rI + \varepsilon$	V vs I: intercept ε , gradient $-r$
$R = \varepsilon(1/I) - r$	R vs 1/I: gradient ε , intercept $-r$
power in cell = I^2r	to load I^2R ; total εI

- Circuit: cell, ammeter, variable resistor in series; voltmeter across the cell. Vary R, plot V vs I.
- Voltmeter reads ε only on open circuit (no current, no p.d. across r).

RESISTIVITY AND DRIFT VELOCITY

$R = \rho l/A$	$\Omega \text{ m}$; strip: $A = \text{width} \times \text{thickness}$
$I = nqvA$	n = carriers per m^3 ; $v \sim 10^{-4} \text{ m s}^{-1}$

- Series: same I, so $v \propto 1/(nA)$: thinner section $\rightarrow$ faster drift. Doubling diameter quarters v.
- Metal heated:** larger lattice vibrations $\rightarrow$ more collisions $\rightarrow$ R up (n barely changes). **Semiconductor:** n increases $\rightarrow$ R down.
- Stretched at constant volume: $l \times 2$, $A \div 2$, so $R \times 4$ ($R \propto l^2$).

RESISTIVITY EXPERIMENT

- Micrometer: diameter at several points/orientations; mean; $A = \pi d^2/4$.
- Wire, ammeter, variable resistor in series; voltmeter across the test length.
- Measure l with a metre rule; $R = V/I$; repeat for several lengths.
- Plot R against l: **$\rho = \text{gradient} \times A$** .
- Small current / switch off between readings so the wire doesn’t heat.

WAVES AND STATIONARY WAVES

$v = f\lambda$; $f = 1/T$	λ from s-x graph; T from s-t graph
phase diff = (path diff/ λ) $\times 2\pi$	$\lambda/2$ or $1.5\lambda \rightarrow$ antiphase (π)
$v = \sqrt{T/\mu}$	μ = mass per unit length; $f \propto \sqrt{T}$
$I = P/A$; point source $A = 4\pi r^2$	inverse square; panel at angle: $P = IA \cos \theta$

- Transverse:** oscillations perpendicular to energy transfer (EM, strings). **Longitudinal:** parallel (sound); zero displacement at centre of compression/rarefaction.
- Coherent:** same f and constant phase difference.
- Diffraction:** spreading through a gap or round an obstacle; greatest when gap $\approx \lambda$.

STATIONARY WAVE FORMATION

Wave reflects at the end; incident and reflected waves (same f , opposite directions) superpose.

Nodes: destructive, zero amplitude. Antinodes: constructive, maximum amplitude.

Node-node = $\lambda/2$; node-antinode = $\lambda/4$. String fundamental $\lambda = 2L$; closed tube $\lambda = 4L$; open tube $\lambda = 2L$.

Points between adjacent nodes in phase; adjacent loops in antiphase. No net energy transfer.

Higher pitch: shorter string, more tension, smaller μ . End correction $\rightarrow$ calculated v too low.

INTERFERENCE AND DIFFRACTION GRATING

- **Maximum:** path difference $n\lambda \rightarrow$ in phase $\rightarrow$ constructive. **Minimum:** $(n + \frac{1}{2})\lambda \rightarrow$ antiphase $\rightarrow$ destructive.
- Speed of sound (two speakers): find a minimum, path difference = $\lambda/2$, $v = f\lambda$. Use microphone + oscilloscope, not ears.

$n\lambda = d \sin \theta$	$d = 1/(\text{lines per m}); 300 \text{ mm}^{-1} \rightarrow d = 3.33 \times 10^{-6} \text{ m}$
$\tan \theta = y/D$	y from central maximum; D to screen
$n_{\text{max}} = d/\lambda$	round down; $\sin \theta > 1 \rightarrow$ order doesn't exist

- Double-slit equations score nothing. White light: violet nearest centre, red furthest.
- Lower uncertainty: measure between first orders either side and halve; larger D ; higher orders.

REFRACTION, POLARISATION, ULTRASOUND

$n_1 \sin \theta_1 = n_2 \sin \theta_2$; $n = c/v$	angles from the normal
$\sin C = 1/n$	general: $\sin C = n_2/n_1$

- **TIR:** higher $n \rightarrow$ lower n and $i > C$. Check: $n_1 \sin \theta_1/n_2 > 1 \rightarrow$ no refracted ray.
- **Refraction explained:** wave slows in denser medium; one part of the wavefront slows first, so the direction changes. f unchanged; v , λ decrease.
- Violet has larger $n \rightarrow$ refracts more. Fibres: different path lengths $\rightarrow$ pulse spreads; cladding protects core.
- **Plane polarised:** oscillations in one plane including the direction of travel. Only transverse waves $\rightarrow$ proof light is transverse.
- One filter rotated: unpolarised $\rightarrow$ constant intensity; polarised $\rightarrow$ max to min every 90° . Crossed filters $\rightarrow$ nothing.
- Reflected light partially polarised; vertical filter (sunglasses) absorbs it. Rotating aerial varies signal $\rightarrow$ radio waves polarised.

ULTRASOUND PULSE-ECHO

1. Pulses sent into the material.
2. Partially reflected at a boundary between media.
3. Time between sending and receiving the echo measured.
4. Speed in the material known.
5. Distance = speed $\times$ time,
6. using **half** the time (there and back).

- Higher $f \rightarrow$ shorter $\lambda \rightarrow$ less diffraction $\rightarrow$ finer detail.

PHOTONS AND ELECTRONS

$E = hf = hc/\lambda$	$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$; photons/s = $P \div E$
$hf = \Phi + \frac{1}{2}mv_{\text{max}}^2$	$f_0 = \Phi/h$
$E_{k,\text{max}}$ vs f	gradient h ; x-int f_0 ; y-int $-\Phi$
$\lambda = h/p = h/mv$	$eV = \frac{1}{2}mv^2 \rightarrow v = \sqrt{(2eV/m)}$

- **Work function:** minimum energy to release an electron from the metal surface.
- **Absorption:** photon needs **exact** level difference; an electron can give part of its KE.
- Lines from level 3: $3 \rightarrow 2$, $3 \rightarrow 1$, $2 \rightarrow 1 = 3$ lines. Largest gap $\rightarrow$ highest f , shortest λ .
- Constants: $h 6.63 \times 10^{-34} \text{ J s}$; $c 3.00 \times 10^8 \text{ m s}^{-1}$; $m_e 9.11 \times 10^{-31} \text{ kg}$; sound $\approx 340 \text{ m s}^{-1}$.
- **Electron diffraction:** rings show wave behaviour; $\lambda \approx$ atomic spacing in graphite. Higher p.d. $\rightarrow$ shorter $\lambda \rightarrow$ smaller rings.

THE 6-MARKERS TO MEMORISE

A • Photoelectric effect vs wave model

1. $E = hf$; one photon interacts with one electron
2. threshold f : photon energy $\geq \Phi$ (wave: any f eventually)
3. emission instantaneous (wave: delay while energy builds)
4. KE_{max} depends on f : $hf = \Phi + KE_{\text{max}}$
5. wave: more intense light would give more KE
6. intensity only changes photons/s, so electrons/s (current)

B • Thermistor in a potential divider

1. T rises, electrons gain energy
2. more conduction electrons released
3. thermistor R decreases
4. in series, so they share the supply p.d.
5. thermistor is a smaller share of total R (I up, V_{fixed} up)
6. p.d. across thermistor decreases

C • Line spectra from a discharge tube

1. p.d. causes current; energy to gas atoms
2. electrons excited to higher levels
3. fall back, emitting a photon
4. photon energy = level difference, $E = hf$
5. levels are discrete, so only certain differences
6. only certain $f/\lambda \rightarrow$ line spectrum

D • Interference from two gaps

1. waves diffract through each gap
2. waves overlap and superpose
3. at X path difference zero, so in phase
4. constructive $\rightarrow$ maximum (loud)
5. moving away, path difference changes
6. $\lambda/2 \rightarrow$ antiphase $\rightarrow$ destructive minimum

E • Stationary wave in a closed tube

1. tuning fork makes sound waves in the tube
2. wave reflects from the closed end
3. reflected and incident waves superpose
4. in phase $\rightarrow$ antinodes; antiphase $\rightarrow$ nodes
5. node at the closed end
6. antinode at the open end, so shortest length = $\lambda/4$

MCQ TRAPS

- Polarisation proves transverse; diffraction, refraction, superposition happen to all waves.
- $v \propto \sqrt{T}$: same string, same T , half length $\rightarrow$ same v .
- n in $I = nqvA$ is per m^3 . Electron λ uses m_e , not the proton mass.
- LDR in divider, more light: p.d. across LDR falls, across fixed resistor rises.